

Figures determining Steps & Result

1. Get your gene:

Choose your target gene i.e. PCSK9

2. Visualization using SnapGene Viewer:



3. Designing Guides using CHOPCHOP:

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CHOPCHOP



Target

Paste fasta entry here.

In

[Add](#) new species.

Using

[Change default PAM and guide length in Options.](#)

For

[Presets can be adjusted in Options.](#)

Gene Target

Options

Reset Options

Find Target Sites!

🌟 Check out our latest set of tools for genome engineering! 🌟

Find all off-targets - even bulges - real quick!

4. Run & Review possible guide sequences:



5. Scoring & Evaluating the Guides in Excel using required filters:

Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	GCCCCATGCTGACTACATCGAGG	seq:6987	+	60	1	1	0	0	0	75.32
2	TTAATCAGCTCAGCATACCGTGG	seq:5294	+	45	0	1	0	0	0	69.63
3	AAGGGGCGGAGGTATTCTCGAGG	seq:1131	+	60	1	1	0	0	0	68.66
9	ATCCACGAGAACCGTAAGGGG	seq:7899	-	55	0	1	0	0	0	58.36

6. Copy the Guides from Excel and view it in SnapGene Viewer.

➤ *Guide_RNA.Cas9*



➤ *Guide_RNA.Cas12a*

This screenshot shows the 'Sequence' view of the PCSK9 gene editing site. The top toolbar includes 'Save', 'Undo', 'Redo', 'Cut', 'Copy', 'Paste', and 'Print'. The main window displays the DNA sequence of PCSK9_gene EDITING.dna (25,305 bp) with a scale from 10 to 100. The sequence is shown in a multi-line format, with positions 15,200 to 15,700 visible. A purple arrow labeled 'Guide_RNA.Cas12a' points to a specific site in the sequence. Below the main sequence, a detailed view of the 'Guide_RNA.Cas12a' sequence is shown, including a green 'PAM' (Protospacer Adjacent Motif) sequence. The sequence is displayed in a multi-line format, with positions 15,200 to 15,700 visible.

➤ *Overall Result View*

This screenshot shows the 'Overall Result View' of the PCSK9 gene editing site. The top toolbar includes 'Save', 'Undo', 'Redo', 'Cut', 'Copy', 'Paste', and 'Print'. The main window displays a genomic map of the PCSK9_gene EDITING.dna (25,305 bp) with a scale from 5,000 to 25,000. The map shows the location of the 'Guide_RNA.Cas9' site (orange), the 'Guide_RNA.Cas12b' site (green), and the 'Guide_RNA.Cas12a' site (purple). The 'PAM' (Protospacer Adjacent Motif) sequence is also indicated. The map is labeled 'PCSK9_gene EDITING' and '25,305 bp'.